LSCS

1-HP Combat System

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1. What is the 1-HP System?

Most strategy games let you pick one of two combat feelings:

- **Standard HP System** – the modern approach. Every hit subtracts a number from a health bar. Predictable, familiar, and it is the default in LSCS.
- **1-HP System** – the classic approach this guide is about. Every soldier basically has **one hit point**. A hit either kills them or it doesn't. Fights last longer, duels feel deadlier, and battles are decided by morale breaking rather than health bars slowly draining.

Both are just assets you can drop into a field. **You can switch between them at any time, and you never have to change your existing unit stats.** See chapter 3 for the one-step swap.

1.1 Why a second system at all?

The Standard system is great for a modern, readable battle where you can see health bars ticking down. The 1-HP system is for when you want that older, tenser feeling: two lines of soldiers grinding against each other for a while, most blows bouncing off shields and armour, until one side's morale finally cracks and they break and run. Neither is "better" – it is a style choice.

2. The Core Idea

This is the **one thing** to understand – everything else in this guide is just fine-tuning around it.

Every time a soldier swings at an enemy, the game secretly flips two coins, one after the other:

1. **Coin 1 – Did it connect?** Did the attack actually get past the enemy's shield, armour and skill? This is a comparison of the attacker's **Attack** value against the defender's **Defence** value.
2. **Coin 2 – Was it fatal?** *Only if* the first coin says yes: was that connecting blow actually a killing blow, or just a scratch, a stagger, a parry that still landed? This depends on the **weapon** – a sword is more likely to finish someone off than a spear-poke.

A soldier only dies if both coins come up in the attacker's favour. Because that is rare, most swings in a fight do nothing dramatic at all – which is exactly the point. It is what makes a battle feel like a real, tense grind instead of two crowds of paper dolls disappearing in three seconds.

> **Why not just one roll?** If you use a single roll for "kill or not", evenly matched soldiers kill each other constantly and your battle is over in a few seconds. Splitting it into two separate rolls is what keeps fights slow, tense, and winnable by morale rather than by whoever swings first.

2.1 Why flanking suddenly becomes so powerful

A soldier hit from the side or the back loses the benefit of their shield and their combat skill for that hit – only their armour still protects them. That is normally a modest change. But because the "connect" coin above reacts very strongly to small changes (chapter 2.2 explains why), that modest change is enough to make the kill chance jump noticeably. You get realistic, punishing flanking for free, without the game needing any special "flanking rule" bolted on.

2.2 Technical detail

You never need to read this section to use the system as it exists purely so a curious designer can see what is under the hood.

The chance for coin 1 (connect) does not rise in a straight line with the stat difference – it rises exponentially. In plain terms: at equal Attack and Defence you get the low baseline chance. Every point of advantage doesn't just add a little – it *multiplies* the chance again. Every point of disadvantage divides it. That is why a small edge in stats, or the small edge from flanking, produces a disproportionately large change in outcome.

```
connect chance = baseConnectChance × growthFactor ^ (Attack − Defence)
(then limited between a minimum and a maximum, so nothing is ever 0% or 100%)
```

You will never have to type this formula anywhere. The sliders in chapter 5 shape this curve for you.

3. Turning It On (One Step)

The 1-HP system is a single ScriptableObject asset. Switching to it is a drag-and-drop, nothing more:

1. Open your **DamageModifierSO** asset (the same one used by the Standard system).
2. Drag the **Rome One 1-HP System** asset into its **Active Combat Ruleset** field.

That's it. Every unit, weapon and formation in your game immediately fights using the new rules. No prefabs to edit, no code to touch, and nothing about your existing unit stats has to change.

> **Good to know:** You can swap back to the Standard HP System at any moment the exact same way – just drag the other ruleset asset into the same field. Nothing is destroyed or locked in.

4. Quick Start

1. Right-click in the Project window → **Create → TopsonGames → Combat Rulesets → Rome One 1-HP System**.
2. Drag the new asset into **Active Combat Ruleset** on your DamageModifierSO.
3. Press Play. It works immediately, using the base damage and armour values your units already have.
4. If fights feel too fast or too slow, open the ruleset asset and adjust **Base Connect Chance** and the lethality values – chapter 5 explains exactly what they do, and chapter 7 gives ready-made recipes.

> **You do not need to touch your unit data:** Out of the box, the 1-HP system simply reuses each unit's existing Base Damage as its Attack, and its existing Armor as its Defence. The extra per-unit fields in chapter 6 exist only if you later want more precise control – you can safely ignore them at the start.

5. Settings

This chapter lists every field on the **Rome One 1-HP System** asset, grouped exactly the way they appear in the Inspector. The listed defaults are a good starting point for the classic feel described in this guide.

5.1 Connect Roll – "did the blow even land?"

Setting	Default	What it does
Base Connect Chance	0.10	The connect chance in a perfectly even fight. **This is the main dial for how deadly combat feels overall.** Lower = slower, more survivable battles. Higher = bloodier, faster battles.
Connect Growth Per Point	1.16	How much each point of stat advantage shifts the odds. 1.16 means roughly +16% for every point of advantage. Higher = stats and flanking swing the fight harder. Lower = stats matter less, fights stay flatter.
Attack Defense Diff Clamp	20	A safety cap on how far the stat gap can push the result, so no matchup is ever a guaranteed kill or a guaranteed miss.
Min Connect Chance	0.01	The floor. Even a hopeless attacker always keeps this tiny chance of landing a hit.
Max Connect Chance	0.95	The ceiling. Even an overwhelming attacker never reaches a certain hit.

5.2 Lethality – "was that hit actually fatal?"

Setting	Default	What it does
Melee Lethality Fallback	0.40	Once a melee blow connects, the chance it is actually a kill rather than a stagger or a parry. Used whenever a weapon has no lethality value of its own (chapter 6). Lower = longer, grindier melee fights.
Ranged Lethality Fallback	0.25	The same idea for arrows and other projectiles. Kept low on purpose, so archers wear an enemy down over time instead of deleting soldiers instantly.

5.3 Hit Points – "one life, or several?"

Setting	Default	What it does
Treat All Units As Single Hitpoint	On	**On:** a fatal hit removes the entire health bar in one go, so every soldier behaves as having exactly one life, no matter what health value it has. You change nothing in your unit data. **Off:** enables true multi-hit-point units – see the box below.

Setting	Default	What it does
Hit Point Damage	1	Only used when the toggle above is **Off**. How much health a single fatal hit removes. Give a unit a health value equal to how many fatal hits it should survive – 1 for a normal soldier, 5 for an elephant.

For elephants, generals and other tough units: Turn *Treat All Units As Single Hitpoint* off, then give just those units a health value of 3–6. They will now shrug off several fatal hits before finally going down – exactly like the classic games handled monsters and heroes. Every normal soldier can simply stay at a health of 1.

5.4 Ranged Combat

Setting	Default	What it does
Ranged Damage Scaling	18	Converts a projectile's damage (after armour is already taken into account) into a connect chance. Higher = arrows connect less often = ranged fire feels weaker. Lower = archers hit harder.

5.5 Armour, Shield & Facing

Setting	Default	What it does
Facing Angle Threshold	60°	The cone in front of a defender in which their shield and combat skill still protect them. Anything hitting from outside this cone counts as a flank or rear attack – the defender keeps only their armour.
Shield Defense Weight	6	Converts a unit's shield block chance (0 to 1) into defence points for the roll. Higher = shields protect the wearer more, but only while facing the attacker.

5.6 Charging & High Ground

These add temporary bonus points to the attacker's Attack value; they do not change the formula itself.

Setting	Default	What it does
Charge Attack Bonus	4	Extra Attack while a formation's charge bonus is active. Makes a charging cavalry or infantry attack noticeably harder-hitting for a few seconds.
Elevation Threshold	1.0 m	The minimum height difference before the high-ground / low-ground bonus below kicks in.
High Ground Attack Bonus	2	Extra Attack when striking from higher ground. Rewards holding the hill.

Setting	Default	What it does
Low Ground Attack Malus	2	Attack penalty when striking uphill.

5.7 Knockback (when a hit is survived)

When coin 1 lands but coin 2 doesn't – a blow connects but isn't fatal – the defender flinches back a little, as if they had parried at the last moment. These three settings control that visual stagger.

Setting	Default	What it does
Knockback Force	0.3	How far a surviving defender is pushed back – the small stagger you see on a parry.
Knockback Sample Range	1.0	A safety margin used when finding a safe spot to push the unit to. Keep it a little larger than the force above.
Knockback Duration	0.25 s	How long that push lasts.

6. Fine-Tuning Individual Units (Optional)

Everything so far works using only a unit's existing **Base Damage** and **Armor** values. If you want more precise control over specific units later, three optional fields are available on each **UnitDataSO**. You can safely leave every one of them at 0 – the system will simply keep using the fallback values from chapter 5 instead.

Setting	Default	What it does
Melee Attack Skill	0	This unit's Attack value for the 1-HP formula. Leave at 0 to reuse the unit's Base Damage automatically.
Melee Defense Skill	0	This unit's Defence skill. Only counts when the unit is hit from the front – flanking removes it, though armour always still counts.
Melee Weapon Lethality	0	The kill chance of this unit's specific weapon once a blow connects. A sword might sit around 0.45, a spear lower at around 0.25. Leave at 0 to use the global fallback from chapter 5.2.

Why bother with per-weapon lethality? Attack and Defence already come from a unit's stats, but nothing else in the system says "a sword is scarier than a spear-poke once it lands." Filling in Melee Weapon Lethality per unit is the simplest way to make different weapons feel genuinely different from each other.

7. Tuning Recipes

Common goals, and the fastest dial to turn for each one. Change **one setting at a time** and test – the curve in chapter 2 means small changes can have a bigger effect than you expect.

Everything dies too fast

1. Lower **Base Connect Chance** first, try 0.06–0.08.
2. If it is still too fast, lower the lethality values too.
3. Also try increasing the attack cooldown on the unit's Combat Behaviour, so soldiers simply swing less often.

Battles never end, units feel unkillable

1. Raise **Base Connect Chance** and/or the lethality values.
2. Or shorten the attack cooldown on the Combat Behaviour so soldiers swing more often.

I want a few tough units – elephants, generals, monsters

1. Turn **Treat All Units As Single Hitpoint** off.
2. Give just those units a health value of 3 to 6.
3. Leave every normal soldier's health at 1.

Archers are too deadly

1. Raise **Ranged Damage Scaling**.
2. And/or lower **Ranged Lethality Fallback**.

I want weapons to feel different from each other

1. Set **Melee Weapon Lethality** per unit – for example sword 0.45, axe 0.50, spear 0.25.

Unit stats (and flanking) should matter more, or less

1. Raise **Connect Growth Per Point** to make good stats and flanking swing a fight harder.
2. Lower it to make the outcome depend less on stats and more on chance.

> **The one rule to remember: Kill rate = chance per swing ÷ time between swings.** This ruleset only controls the chance per swing. How *often* a unit swings is a completely separate setting – the attack cooldown on the unit's Combat Behaviour. If a fight still feels wrong after adjusting the ruleset, that cooldown is the other half of the answer.

8. A Example

Using the default settings from chapter 5 (base connect chance 0.10, growth 1.16, melee lethality 0.40). The final kill chance for one swing is simply **connect chance × lethality chance**.

Situation	Connect chance	Lethality	Kill chance per swing
Even fight (Attack = Defence)	0.10	0.40	~4%
Attacker is 5 points stronger	~0.21	0.40	~8%
Defender is flanked (loses shield + skill)	~0.24	0.40	~10%

Notice how an even fight is deliberately slow – only about a 4% chance to actually kill on any given swing – while a real stat advantage or a flank attack pushes that noticeably higher. That is the whole feel of the classic system in one table: two lines of soldiers trade blows for a while without much happening, and then the fight tips quickly and decisively the moment one side gets an edge – usually right around the same time the losing side's morale starts to break.

9. Answers to Questions

Do I have to change my unit data to use this?

No. It runs on your existing Base Damage and Armor values automatically. The optional per-unit fields in chapter 6 are only there for extra control later.

Can I switch back to the modern HP system afterwards?

Yes, at any time. Just drag the Standard HP ruleset back into the **Active Combat Ruleset** field on your DamageModifierSO – see chapter 3.

Why do so many hits not kill anything?

That is intentional. The two-roll model in chapter 2 means most blows are parried, blocked or simply glance off. It is exactly what makes fights feel slow and tense, and lets morale decide the outcome instead of raw kill speed.

Where do a unit's Attack and Defence actually come from?

By default: Attack comes from the unit's Base Damage, and Defence comes from its Armor. You can override both with the optional Melee Attack Skill and Melee Defense Skill fields from chapter 6 if you want more precise control over specific units.

My whole army suddenly turns and runs – is that this system?

No, that is the separate morale system, not the combat ruleset described here. What the 1-HP system does is simply make fights slow enough that morale has time to build up and eventually decide the outcome, instead of the battle being over before morale matters at all.

Thank you for using the 1-HP Combat System. If a setting is still unclear, or you are aiming for a very specific battle feel, start from the defaults in chapter 5 and change one dial at a time – **Base Connect Chance** is the master dial and the best place to begin.